

Linear Regression on Drone Flight Data

Problem Description/Objective:

The goal of this model is to see the effect of different flight conditions impacting power consumption while in flight. The model shall identify which predictor variables significantly influence the response variable (Power).

The project will use drone sensor data from various flights using the Carnegie Mellon drone flight dataset. The system is a quadcopter drone which is flown in varying flight conditions. The onboard drone sensors collected and recorded telemetry data during each flight.

Electrical power is a key performance metric. This model can be used to show how much energy is consumed, how efficient the system is, and how long the system can operate in flight to help optimize flight mission planning or control systems.

When modeling the system, the goal is to develop a multiple linear regression model to predict power consumption. The model can be used to perform hypothesis tests to identify statistically significant predictor variables, show how adequate the model is using R^2 residual testing, and check for assumptions using diagnostic plots.

Data Requirements:

The response variable of our model, Power (Watts), is not given from the dataset but is derived using observations of voltage and current and the relation that power equals voltage times current.

$$P = V \cdot I$$

Where P equals power in Watts, V equals voltage in Volts, and I equals current in Amps. Voltage and current are simply inputs used to derive the response and will be excluded from this model. The predictor variables are selected based on the variables collected from the flight data set, and two qualitative variables are created to capture any variation that the quantitative variables do not capture. Table 1 shows each variable in the model showing its variable type, data type, and units.

Table 1 – Variables with Definitions			
Variables	Variable Type	Data Type	Units
Power	Dependent	Quantitative - Continuous	Watts
Time	Independent	Quantitative - Continuous	seconds
Altitude	Independent	Quantitative - Continuous	m
Speed	Independent	Quantitative - Continuous	m/s
Payload Mass	Independent	Quantitative - Static	lbs.
Position	Independent	Quantitative - Continuous	m
Velocity	Independent	Quantitative - Continuous	m/s
Acceleration	Independent	Quantitative - Continuous	lbs.
Orientation	Independent	Quantitative - Continuous	Degrees
Wind speed	Independent	Quantitative - Continuous	m/s
Wind angle	Independent	Quantitative - Continuous	Degrees
Flight Phase	Independent	Qualitative - Categorical	None
Wind Category	Independent	Qualitative - Categorical	None

Given one response variable and 13 predictor variables, this model shows physically meaningful variables and avoids multicollinearity. The only variable that is time-independent will be the payload mass which remains constant. For the time dependent variables each time stamp of data gives an observation showing these variables have potential correlation. The variables may violate the independence assumption which we can determine during analysis, and the model does not need to include all predictors.

The raw dataset has over 250,000 samples. For this model, we will decrease the sample size to 628. The large raw data set has observations very close together within milliseconds, creating noise in the telemetry data. To solve that issue, the data was parsed to include only 3 observations that have a longer time between them for each flight number. This results in a clean data set with 628 samples and improves the model by giving a more stable point in the physics for each observation. The data points include the 3 most important phases of each flight: takeoff, cruise, and landing.

Citations

Rodrigues, T. A., Patrikar, J., Choudhry, A., Feldgoise, J., Arcot, V., Gahlaut, A., Lau, S., Moon, B., Wagner, B., Matthews, H. S., Scherer, S., & Samaras, C. (2021). *In-flight positional and energy use data set of a DJI Matrice 100 quadcopter for small package delivery*. **Scientific Data**, 8. <https://www.nature.com/articles/s41597-021-00930-x>

Link to the raw data set:

https://kilthub.cmu.edu/articles/dataset/Data_Collected_with_Package_Delivery_Quadcopter_Drone/12683453?file=26385151